Measuring the verification gap in municipal energy and infrastructure funding: a source-coded register of federal and New York State programs

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Federal and New York State programs that fund municipal energy and infrastructure projects routinely state expected energy, cost, emissions, and resilience benefits. Whether those benefits can later be confirmed depends on three design features that are rarely recorded together: what proof of results a program requires of recipients, what results the program makes public, and whether the resulting data could support attribution of outcomes to the funding. We introduce the Verification Gap Ledger, an open and versioned register that codes programs on three ordinal scales, required verification (R, 0 to 4), public results (C, 0 to 4), and outcome attributability (A, 0 to 3), with every code tied to a cited source document and locator. Version 0.1.0 covers 24 programs (15 federal, 9 New York State) supported by 80 evidence items. In this sample, 11 programs (46 percent) require no more than output reporting and 5 (21 percent) require measured or independently verified outcomes; none of those 5 publishes project-level outcome data. In addition, 12 programs (50 percent) publish no results beyond award announcements, 22 (92 percent) are coded at or below a modeled baseline for attributability, and none uses a comparison group or formal impact design. The two programs that publish project-level outcomes report modeled estimates. Required verification and public results are weakly associated (Kendall's $\tau_b = 0.19$, $p = 0.30$), while required verification and attributability are strongly associated ($\tau_b = 0.70$, $p < 0.001$). Where independent measurement exists it diverges from estimates: a billing-analysis evaluation of one New York program found a gross electric realization rate of 20 percent. The central finding, that no program both measures and publishes project-level outcomes, holds under every treatment of low-confidence codes that we test. An illustrative scenario in which programs that already measure outcomes publish them shows how the register can be used to evaluate disclosure policy.

Keywords: measurement and verification, program evaluation, municipal energy, infrastructure finance, performance reporting, transparency, open data

I. INTRODUCTION

Public funding for municipal energy and infrastructure in the United States has expanded substantially since 2021. The Infrastructure Investment and Jobs Act (IIJA) [1] and the Inflation Reduction Act (IRA) [2] created or recapitalized programs for building efficiency, grid resilience, electric vehicle charging, clean school buses, water infrastructure, and hazard mitigation, and opened direct payment of clean energy tax credits to state and local governments. New York State operates a parallel portfolio through the New York State Energy Research and Development Authority (NYSERDA), the Department of Environmental Conservation (NYSDEC), the Environmental Facilities Corporation (EFC), and the New York Power Authority (NYPA). Municipalities are eligible recipients, subrecipients, borrowers, or participants in many of these programs.

These programs are justified in part by expected outcomes: kilowatt hours saved, emissions avoided, outage minutes reduced, losses avoided. The credibility of those expected outcomes depends on what happens after the award. A program can require recipients to report only financial information, to count delivered outputs, to estimate outcomes with deemed or modeled values, to measure outcomes against a baseline, or to submit measurement to an independent party. Separately, a program may publish nothing, publish totals, or publish project-level data that others can analyze. Finally, the data a program collects may or may not support a defensible claim that the funding caused the observed result.

These three properties are distinct, and they need not move together. A program can collect metered data and publish none of it; another can publish detailed project records that contain only modeled estimates. Yet municipal officials choosing among funding sources, auditors reviewing performance claims, and researchers studying program effects have no single public record of how programs differ on these dimensions. Each program's requirements are spread across statutes, regulations, notices of funding opportunity, reporting templates, compliance supplements, program pages, and evaluation reports.

Measurement and verification (M&V) is the established practice for determining the savings of an energy project. The International Performance Measurement and Verification Protocol (IPMVP) defines savings as the difference between measured post-project consumption and an adjusted baseline, with options ranging from partial measurement of retrofitted equipment to whole-facility utility bill analysis [3]. M&V performed by a

party independent of the recipient is the strongest form of verification short of an impact evaluation. Much program reporting relies instead on deemed savings, engineering calculations, or applicant projections. These are useful for planning, but they are estimates, not measurements.

A body of empirical work compares projected and realized savings in publicly supported programs. Allcott and Greenstone argued that the engineering estimates commonly used to justify efficiency programs are an unreliable guide to realized savings and called for more rigorous ex post evaluation [4]. Using a randomized encouragement design in the federal Weatherization Assistance Program, Fowlie, Greenstone, and Wolfram found that projected savings substantially exceeded realized savings [5]. Davis, Fuchs, and Gertler found smaller than expected savings in a large appliance replacement program [6], and Christensen and coauthors decomposed the gap between projected and realized returns into components attributable to engineering models, installation quality, and behavior [7]. A common finding is that the size of the gap is unknown until measurement is performed, and that measurement is often absent from routine program reporting.

Showing that a measured change was caused by funding requires a counterfactual. Measured pre and post comparisons at project level control for some but not all confounders, such as weather, occupancy, and concurrent changes. Comparison group, randomized, or quasi-experimental designs, and net-to-gross adjustments in utility program evaluation, address attribution more directly [8]. The data a program requires therefore determine whether such analysis is possible after the fact: a program that collects no baseline cannot support attribution, regardless of later analytical effort.

Federal performance management law requires agencies to set goals and report on them [9] and to build evidence about program effectiveness [10]. For individual awards, the Uniform Guidance requires periodic performance reports relating accomplishments to award objectives, at least annually, but it recommends rather than requires that awards specify baselines and outcome indicators [11]. Research on performance management has long noted the distance between outputs that are easy to count and outcomes that matter [12, 13]. Research on transparency policy shows that disclosure is effective only when the information disclosed is usable by its intended audience [14, 15]; open, machine-readable release is one condition for usability, consistent with the FAIR principles for research data [16].

This literature evaluates individual programs or describes general reporting obligations. It does not provide a cross-program register that records, for each program and with source citations, what verification is required, what is published, and whether attribution is possible. We contribute that register, the Verification Gap Ledger (VGL) [17]. The approach follows a familiar pattern in applied evaluation, in which a program's stated objectives are compared with what its design can deliver or demonstrate using transparent, reproducible indicators [18]; here the indicators concern evidence rather than access. The Ledger does not estimate program effects. It records the evidentiary conditions under which program effects could be estimated by others.

The paper is organized as follows. In Sec. II we set out the coding framework formally, including the three scales, the derived measures, the association statistic, and the operators used for sensitivity analysis and scenario evaluation. Section III describes the study scope, the program sample, the source hierarchy, the decision rules, and the reproducible pipeline. Section IV reports results. Section V interprets them through a sensitivity analysis, an illustrative disclosure scenario, and policy implications. Section VI states limitations, and Sec. VII concludes and sets out the release schedule.

II. THE VERIFICATION GAP FRAMEWORK

This section defines the quantities used throughout the paper. Let $\mathcal{P}$ denote the set of coded programs, with $n = |\mathcal{P}| = 24$ in version 0.1.0. Each program $i \in \mathcal{P}$ receives one code on each of three ordinal scales,

$$R_i \in \{0,\dots,4\}, \qquad C_i \in \{0,\dots,4\}, \qquad A_i \in \{0,\dots,3\}, \tag{1}$$

where R_i records the verification of results that the program's rules require of recipients, C_i records the results that any member of the public can see, and A_i records whether the required or published data could support attribution of outcomes to the funding. The levels of each scale and the test that assigns them are given in Table I.

A. Derived measures

Two derived measures summarize each program. The required-to-public gap is

$$G_i = R_i - C_i, \tag{2}$$

which is positive when the rules ask for more proof than is published and negative when more is published than the rules require. The outcome gap is

$$O_i = 4 - \max(R_i, C_i), \tag{3}$$

which measures distance from the highest evidence level reached on either the requirement or the disclosure scale, with $O_i = 0$ best. Both measures subtract ordinal codes for convenience. A difference of 2 is not interpreted as twice a difference of 1, and means of codes are reported only as summaries.

Headline indicators are counts and shares of programs satisfying a condition ϕ,

$$N(\phi) = \sum_{i \in \mathcal{P}} \mathbf{1}[\phi(i)], \qquad s(\phi) = \frac{N(\phi)}{n}, \tag{4}$$

TABLE I: Coding scales of the Verification Gap Ledger. Source: docs/CODEBOOK.md, Verification Gap Ledger v0.1.0.

Scale	Code	Label	Test
R, required verification	0	Financial only	Only financial reports or eligibility documentation required
	1	Outputs	Recipients report activities or units delivered
	2	Estimated outcomes	Outcome quantities are modeled, deemed, projected, or calculated
	3	Measured outcomes against a baseline	Outcomes from meters, bills, or operating records, against a baseline or measured operating performance
	4	Independent verification	M&V or impact evaluation by a party independent of the recipient
C, public results	0	No public results	No public release of results; award announcements and recipient lists do not count
	1	Aggregate only	Program totals or evaluation summaries only
	2	Project-level outputs	Outputs published per project or recipient
	3	Project-level outcomes	Outcome quantities (estimated or measured) published per project
	4	Machine-readable open project data	Project-level outcome fields as bulk machine-readable data with open terms
A, outcome attributability	0	No baseline	No baseline required or published
	1	Baseline or modeled counterfactual only	A baseline exists but results are not measured against it, or the counterfactual is modeled
	2	Measured pre and post	Measured outcomes before and after, at project level
	3	Comparison group or impact evaluation	Comparison group, randomized or quasi-experimental design, or net-to-gross impact evaluation

for the five conditions $R_i \leq 1$ (outputs or less), $R_i \geq 3$ (measured or independently verified), $C_i = 0$ (no public results), $A_i \leq 1$ (at or below a modeled baseline), and the joint condition $R_i \geq 3 \wedge C_i \geq 3$, which identifies programs that both measure and publish project-level outcomes. The count $N(G_i > 0)$ of programs with a positive gap and the mean outcome gap $\bar{O} = n^{-1} \sum_i O_i$ complete the set.

B. Evidence and confidence

Let $\mathcal{E}_i^d$ be the set of evidence items recorded for program i on dimension $d \in \{R, C, A\}$, and let each item e carry a code $c(e)$ and a confidence $q(e) \in \{\text{low} < \text{medium} < \text{high}\}$: high when the rule text was read directly, medium for an official secondary description or partial text, and low when the code is inferred from incomplete official material. Every assigned code must be supported by at least one item, and the confidence of a program on a dimension is the strongest support for the code assigned,

$$\kappa_i^d = \max \left\{ q(e) : e \in \mathcal{E}_i^d,\ c(e) = d_i \right\}. \tag{5}$$

The release pipeline rejects any program for which this set is empty.

C. Association between scales

Because the codes are ordinal with many ties, associations are summarized with Kendall's τ_b [19]. For two scales X and Y observed on the same n programs,

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_X)(n_0 - n_Y)}}, \tag{6}$$

where n_c and n_d are the numbers of concordant and discordant pairs, $n_0 = n(n-1)/2$, and $n_X = \sum_k t_k(t_k - 1)/2$ and $n_Y = \sum_l u_l(u_l - 1)/2$ count pairs tied on X and on Y, with t_k and u_l the sizes of the tied groups. Two-sided p-values use the asymptotic normal approximation as implemented in SciPy [20]. Given the small and purposive sample, the coefficients are descriptive.

D. Realization rate

For a program with an independent evaluation, the gross realization rate is the ratio of evaluated to estimated savings,

$$\rho = \frac{S_{\text{measured}}}{S_{\text{estimated}}}, \tag{7}$$

so that $\rho = 1$ means estimates were confirmed. The Ledger records ρ only where a cited evaluation reports it.

E. Sensitivity and scenario operators

To test how far findings depend on low-confidence codes, we define four alternative treatments. Two restrict the sample,

$$\mathcal{P}_a = \{i : \kappa_i^R \neq \text{low}\}, \qquad \mathcal{P}_b = \{i : \kappa_i^d \neq \text{low}\ \forall d\}, \tag{8}$$

and two shift every low-confidence code by $\delta = \pm 1$ while keeping it inside its scale,

$$\tilde{d}_i = \min\big(d_{\max}, \max(0, d_i + \delta\, \mathbf{1}[\kappa_i^d = \text{low}])\big), \tag{9}$$

with $d_{\max} = 4$ for R and C and 3 for A. Treatment (c) uses $\delta = +1$ and bounds the case in which unreviewed award terms are more demanding than the evidence shows; treatment (d) uses $\delta = -1$.

Finally, a disclosure scenario raises public results to the project-outcome level for every program that already

requires measured or independently verified outcomes, and changes nothing else:

$$C_i' = \begin{cases} \max(C_i, 3) & \text{if } R_i \geq 3, \\ C_i & \text{otherwise.} \end{cases} \tag{10}$$

Two properties follow directly. Under Eq. (10) the gap of an affected program becomes $G_i' = R_i - \max(C_i, 3)$, which is zero for $R_i = 3$ and one for $R_i = 4$. The outcome gap is unchanged, because for every affected program $\max(R_i, C_i') = R_i = \max(R_i, C_i)$, so $O_i' = O_i$. We return to the second property in Sec. B.

III. DATA AND CODING PROTOCOL

A. The federal reporting floor

All federal financial assistance is subject to the Uniform Guidance in 2 CFR Part 200 [11]. Two provisions set the minimum for performance evidence. Section 200.329 requires recipients to submit performance reports that relate accomplishments to award objectives, no less frequently than annually, with a final report within 120 days of the end of the period of performance. Section 200.301 directs agencies to measure performance and recommends that awards specify outcomes, indicators, targets, and baseline data. Neither provision requires measured outcomes or independent verification. Program-specific statutes, regulations, and guidance therefore determine whether a program rises above the floor. The Ledger records these program-specific requirements and treats the Uniform Guidance as a cross-cutting reference rather than as evidence for any single code.

B. New York State context

New York funds municipal energy work through several channels. NYSERDA administers programs under the Clean Energy Fund, authorized by the Public Service Commission in Case 14-M-0094. NYSDEC administers Environmental Protection Fund grants and rebates, including Climate Smart Communities grants and municipal zero-emission vehicle programs. EFC administers the Green Innovation Grant Program through the Clean Water State Revolving Fund, and NYPA delivers turnkey street lighting conversions. A July 2024 report of the Office of the State Comptroller reported Clean Energy Fund progress as acquired versus committed savings without assessing whether reported savings were estimated or independently verified [21]. New York therefore offers a useful comparison with federal programs: it maintains an established program evaluation function, and some of its programs have been subject to third-party impact evaluation.

C. Programs in version 0.1.0

Version 0.1.0 covers 24 programs, 15 federal and 9 New York State, listed in Table II. They were selected to span administering agencies, funding vehicles (formula grants, competitive grants, revolving loans, rebates, and tax credit payments), and asset classes (buildings, street lighting, water and wastewater, fleets and charging, grid, and hazard mitigation). The sample is purposive, not random or complete.

D. Program volatility during the coding period

Four of the 24 programs changed status materially in 2025 and 2026. EPA announced on August 7, 2025 that it would no longer implement Solar for All [44]; on September 18, 2026 the U.S. District Court for the District of Rhode Island vacated the termination [45]. FEMA announced the end of BRIC on April 4, 2025, and on December 11, 2025 a federal court ruled the termination unlawful and permanently enjoined it [46]. The NEVI Formula Program was suspended in February 2025, reissued under revised guidance in August 2025, partly rescinded in January 2026, and had its spacing requirements removed in February 2026 [47]. Federal tax legislation in 2025 changed eligibility timelines for several clean electricity credits available through elective pay. These programs remain in the Ledger with their status recorded, because the coding protocol keeps terminated programs on record: the evidence a program generated does not lapse when its funding does.

E. Inclusion rule and unit of analysis

A program is in scope when three conditions hold: it is administered by a federal agency or a New York State agency or authority; municipalities, including cities, towns, villages, counties, school districts, municipal utilities, and local public authorities, can receive its funds or benefits directly, as subrecipients, or as borrowers; and its eligible uses include energy, energy-dependent infrastructure, or infrastructure resilience. The unit of analysis is the program as the funder defines it. Where a portfolio contains named programs, such as the NYSERDA Clean Energy Fund, the named programs are coded rather than the portfolio, to avoid double counting. Ledger identifiers (VGL-F-### for federal and VGL-NY-### for New York State programs) are permanent and never reused.

F. Sources and source hierarchy

Each code is assigned from the highest available tier of evidence: (1) statute, regulation, or award terms and conditions; (2) funding notice, program guidance, reporting templates, or the OMB Compliance Supplement; (3) official program pages, agency data catalogs, and evaluation reports; and (4) independent oversight bodies and reputable secondary sources, used for status and context, and for C only where they describe what is published. When sources conflict, the higher tier governs and both are recorded. Version 0.1.0 draws on documents from DOE, EPA, DOT (FHWA and FTA), HUD, USDA, FEMA, Treasury and IRS, NYSERDA, NYSDEC, NYS EFC, NYPA, GAO, the Congressional Research Ser-

TABLE II: Programs coded in Verification Gap Ledger v0.1.0. ALN is the federal Assistance Listing Number. The last column cites the primary source for each program's R code. Status reflects sources accessed in October 2026.

Ledger ID	Program	Agency	ALN	Funding vehicle	Status	Source
VGL-F-001	Energy Efficiency and Conservation Block Grant Program (IIJA)	DOE	81.128	formula and competitive grant	Active	[22]
VGL-F-002	Grid Resilience State and Tribal Formula Grants (IIJA 40101(d))	DOE	81.254	formula grant to states, subawards	Active	[23]
VGL-F-003	Clean Water State Revolving Fund	EPA	66.458	capitalization grant to state revolving loan fund	Active	[24]
VGL-F-004	Drinking Water State Revolving Fund	EPA	66.468	capitalization grant to state revolving loan fund	Active	[25]
VGL-F-005	Solar for All (Greenhouse Gas Reduction Fund)	EPA	66.959	competitive grant to states, municipalities, tribes, nonprofits	Terminated, contested	[26]
VGL-F-006	Congestion Mitigation and Air Quality Improvement Program	DOT (FHWA, FTA)	20.205	formula apportionment to states	Active	[27]
VGL-F-007	National Electric Vehicle Infrastructure Formula Program	DOT (FHWA)	20.205	formula apportionment to states	Active, modified	[28]
VGL-F-008	Charging and Fueling Infrastructure Discretionary Grant Program	DOT (FHWA)	20.205	competitive grant	Active	[28]
VGL-F-009	Elective Pay for clean energy tax credits (IRA sec. 6417)	Treasury (IRS)	–	refundable tax credit payment	Active, modified	[29]
VGL-F-010	Low or No Emission Vehicle Program	DOT (FTA)	20.526	competitive grant	Active	[30]
VGL-F-011	Clean School Bus Program	EPA	66.045	rebates and competitive grants	Active	[31]
VGL-F-012	Building Resilient Infrastructure and Communities	FEMA	97.047	competitive grant via state, subapplications	Terminated, contested	[32]
VGL-F-013	Community Development Block Grant, Entitlement Communities	HUD	14.218	formula grant	Active	[33]
VGL-F-014	Water and Waste Disposal Loan and Grant Program	USDA (RD)	10.760	direct loan and grant	Active	[34]
VGL-F-015	Climate Pollution Reduction Grants, Implementation	EPA	66.046	competitive grant	Active	[35]
VGL-NY-001	Clean Energy Communities	NYSERDA	–	action-based grants to municipalities	Active	[36]
VGL-NY-002	Real Time Energy Management	NYSERDA	–	cost-share incentive	Active	[37]
VGL-NY-003	Clean Green Schools Initiative	NYSERDA	–	competitive grant	Closed	[38]
VGL-NY-004	Charge Ready NY 2.0	NYSERDA	–	per-port rebate	Active	[39]
VGL-NY-005	Climate Smart Communities Grant Program	NYSDEC	–	50/50 matching grant	Active	[40]
VGL-NY-006	Municipal Zero-Emission Vehicle Rebate Program	NYSDEC	–	rebate	Active	[41]
VGL-NY-007	Municipal Zero-Emission Vehicle Infrastructure Grant Program	NYSDEC	–	competitive grant	Active	[40]
VGL-NY-008	Smart Street Lighting NY	NYPA	–	turnkey project with financing	Active	[42]
VGL-NY-009	Green Innovation Grant Program	NYS EFC	–	competitive grant	Active	[43]

vice, and the NYS Office of the State Comptroller. All 80 evidence items were accessed in October 2026 and are recorded with URL, locator (section, page, slide, or field), access date, and a paraphrased finding.

G. Decision rules

Four rules govern coding. First, R codes obligation, not practice: it records what recipients must report. Where an estimate is required only at application, for example a benefit cost ratio, $R = 2$ is assigned and the evidence note says so. Second, C codes publication, not collection: it records what any member of the public can see. Third, A codes design: whether the required or published data could support attribution, independent of whether anyone has performed that analysis. Fourth, when evidence is missing, the lowest code that the available evidence supports is assigned and confidence is marked low. Notes are paraphrased, and quotation is limited to short phrases where exact wording matters.

H. Verification and reliability

Version 0.1.0 is single-coder. Every evidence item was confirmed by the maintainer against its cited source before release, and the release gate requires zero open verification items. From version 0.5, a second coder will independently code a random 20 percent sample without seeing the first codes, and agreement will be reported as percent agreement and weighted kappa for each dimension [48].

I. Reproducible pipeline

Coded entries are stored as structured JSON. A build script (01_build.py) validates identifiers and code ranges, checks that every code is backed by at least one evidence item with a URL, and writes the processed tables (vgl_programs.csv, vgl_evidence.csv, vgl_reference_sources.csv), a quality report, and stats.json. A second script (05_preprint_stats.py) computes the additional quantities reported in Secs. IV and V, namely the means, the τ_b coefficients, the level and confidence breakdowns, the sensitivity treatments, and the disclosure scenario, and writes stats_preprint.json. A third script (06_preprint_assets.py) produces every figure, every data table, and every number in this paper from those files. Processed tables are never edited by hand. Data and documents are licensed CC BY 4.0 and

TABLE III: Number of programs at each code level, and mean code, for the three scales ($n = 24$).

Scale	0	1	2	3	4	Mean
R, required verification	2	9	8	3	2	1.75
C, public results	12	5	5	1	1	0.92
A, attributability	16	6	2	0	–	0.42

TABLE IV: Kendall's τ_b between pairs of scales, Eq. (6), with two-sided asymptotic p-values.

Pair	τ_b	p	n
R and C	0.19	0.295	24
R and A	0.70	< 0.001	24
C and A	0.26	0.162	24

code MIT.

IV. RESULTS

A. Distribution of the three scales

Figure 1 and Table III summarize the distribution of codes. Required verification is concentrated at the low and middle levels: 2 programs require financial reporting only ($R = 0$), 9 require output reporting ($R = 1$), 8 require estimated outcomes ($R = 2$), 3 require measured outcomes against a baseline ($R = 3$), and 2 are coded at independent verification ($R = 4$). In total, 11 programs (46 percent) require no more than outputs, and 5 (21 percent) require measured or independently verified outcomes.

Public results are lower than required verification. Of the 24 programs, 12 (50 percent) publish no results beyond award announcements or recipient lists ($C = 0$). Of the remainder, 5 publish aggregate results only, 5 publish project-level outputs, 1 publishes project-level outcomes, and 1 publishes project-level outcome fields as bulk machine-readable open data. The mean C code is 0.92, against a mean R code of 1.75.

Attributability is the weakest dimension. A total of 16 programs (67 percent) have no required or published baseline ($A = 0$), 6 have a baseline or modeled counterfactual only, and 2 have measured pre and post outcomes at project level. No program is coded $A = 3$. In total, 22 programs (92 percent) are coded $A \leq 1$, and the mean A code is 0.42. Codes therefore fall from the rules programs set (R), to what they publish (C), to what could support attribution (A).

B. Required verification against public results

Figure 2 places each program by its R and C codes, and Fig. 3 shows the gap G_i of Eq. (2) for every program. A total of 14 programs (58 percent) have a positive gap, meaning their rules require more proof than they publish; 5 have a negative gap and 5 a gap of zero. The mean outcome gap of Eq. (3) is $\bar{O} = 2.00$ on a scale where 0 is best.

The two quadrants that matter most are empty. No program that requires measured or independently verified outcomes ($R \geq 3$) publishes project-level outcomes ($C \geq 3$). Conversely, the two programs that do publish project-level outcomes, the Congestion Mitigation and Air Quality Improvement Program ($R = 2$, $C = 3$) and Building Resilient Infrastructure and Communities ($R = 2$, $C = 4$), publish modeled estimates: ex ante emissions reductions against a no-build case for CMAQ, and benefit cost ratios against a modeled no-action baseline for BRIC [27, 32]. In this sample, the programs that publish outcomes do not measure them, and the programs that measure outcomes do not publish them at project level.

The five programs coded $R \geq 3$ illustrate the pattern. NEVI and the Charging and Fueling Infrastructure grants require quarterly metered operating data per charging port under 23 CFR 680.112, but the regulation permits public release only in aggregated and anonymized form, and a national aggregate release had not been confirmed at the time of coding ($C = 1$) [28]. The Grid Resilience formula grants require reliability indices such as SAIDI and SAIFI measured before and after implementation, but no public release of subrecipient impact data was located ($C = 0$) [23]. NYSERDA's Clean Energy Communities and Real Time Energy Management programs have been subject to third-party impact evaluations whose results are public at program level only ($C = 2$ and $C = 1$) [36, 37].

Programs with a negative gap publish more than their rules require. They are concentrated among programs served by agency data systems: the EPA State Revolving Fund public portal [24], the FHWA CMAQ Public Access System [27], OpenFEMA [32], and the New York open data portal for Charge Ready NY [39]. Disclosure in these cases is a property of the agency's data infrastructure rather than of the reporting obligation placed on recipients.

C. Association among the scales

Table IV and Fig. 4 report Kendall's τ_b of Eq. (6) for each pair of scales. Required verification and public results are weakly associated ($\tau_b = 0.19$, $p = 0.30$). Required verification and attributability are strongly associated ($\tau_b = 0.70$, $p < 0.001$), which is expected because a measured baseline is a precondition for both $R = 3$ and $A = 2$. Public results and attributability are weakly associated ($\tau_b = 0.26$, $p = 0.16$). These coefficients indicate that the evidentiary strength of a program's rules is a poor guide to what it discloses.

D. Federal and state programs

The federal and state subsamples show similar patterns (Table V, Fig. 5). State programs have a higher mean required verification code, driven by the two NYSERDA programs with independent evaluations. Federal pro-

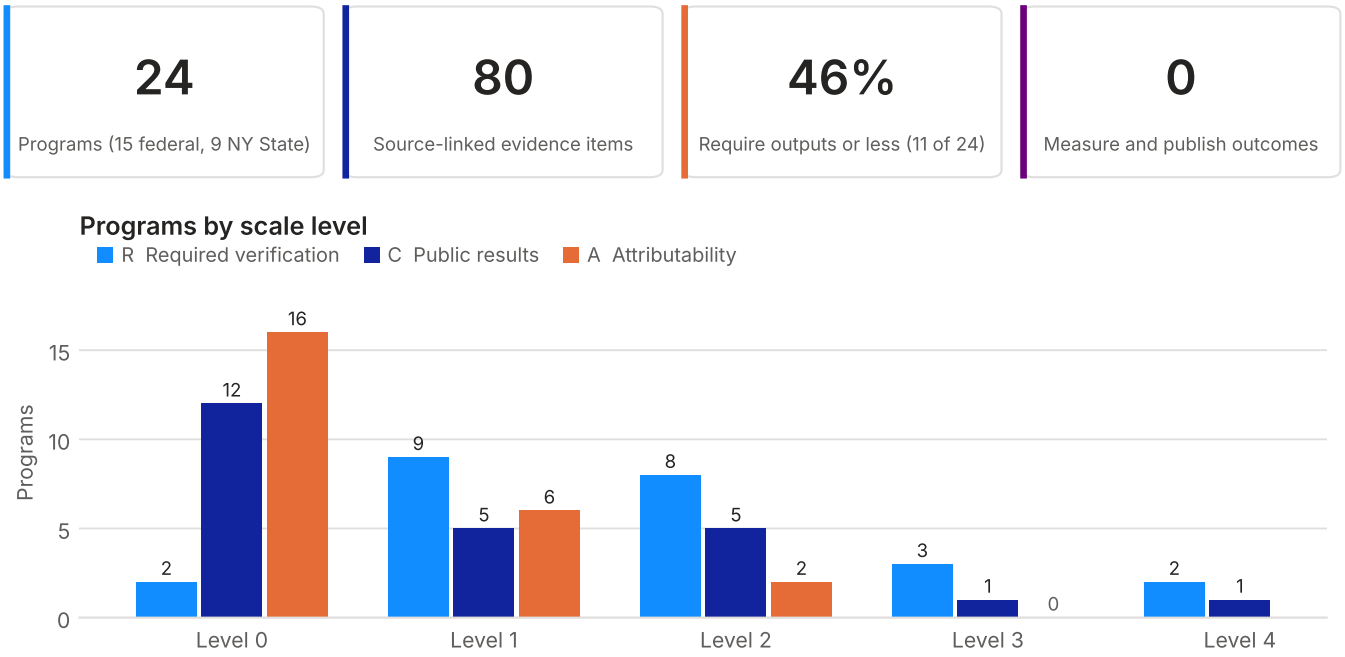


FIG. 1: Overview of Verification Gap Ledger v0.1.0. Cards report the size of the register and two headline indicators; the clustered columns show the number of programs at each level of the three scales. The attributability scale has no level 4. No program both requires measured or independently verified outcomes and publishes project-level outcomes.

TABLE V: Summary indicators by level of government. Counts are numbers of programs; G_i and $\bar{O}$ follow Eqs. (2) and (3).

Indicator	Federal ($n = 15$)	NY State ($n = 9$)	All ($n = 24$)
Mean R	1.60	2.00	1.75
Mean C	1.00	0.78	0.92
Mean A	0.47	0.33	0.42
$R \leq 1$ (outputs or less)	7	4	11
$R \geq 3$ (measured or verified)	3	2	5
$C = 0$ (no public results)	7	5	12
$A \leq 1$	14	8	22
$R \geq 3$ and $C \geq 3$	0	0	0
Positive gap $G_i > 0$	7	7	14
Mean outcome gap $\bar{O}$	2.07	1.89	2.00

grams publish slightly more, driven by the OpenFEMA and CMAQ public data systems and the State Revolving Fund portal. Neither subsample contains a program that both measures and publishes project-level outcomes.

E. Evidence confidence

Confidence is uneven across dimensions (Table VI, Fig. 6). Required verification codes rest on low-confidence evidence for 9 programs. Low-confidence R codes arise mainly where a program page defers reporting rules to award terms or application packages that were not reviewed. Public results codes are mostly medium confidence, because absence of publication is established by search of official channels rather than by a rule text. In total, 9 R codes, 5 C codes, and 8 A codes rest on low-confidence evidence; Sec. A tests which findings depend on them.

TABLE VI: Programs by confidence κ_i^d of the evidence supporting each code, Eq. (5).

Dimension	High	Medium	Low
R, required verification	6	9	9
C, public results	2	17	5
A, attributability	3	13	8

F. What independent measurement shows

The one program in the sample with a published billing-analysis evaluation shows why the distinction between estimated and measured outcomes matters. An independent impact evaluation of NYSERDA's Real Time Energy Management program applied IPMVP Option C whole-building billing analysis to 39 of 42 sampled sites and reported gross realization rates, Eq. (7), of $\rho = 20$ percent for electricity, 42 percent for fossil fuels, and 24 percent combined; evaluated savings were about 3 percent of baseline use [37]. Government facilities accounted for 17 of 293 projects and K-12 schools for 14. The evaluation measured pre and post consumption but did not use a comparison group or address net-to-gross attribution ($A = 2$). Figure 7 shows the three rates against the estimate.

A single evaluation does not establish a general realization rate. It shows that reported estimates and mea-

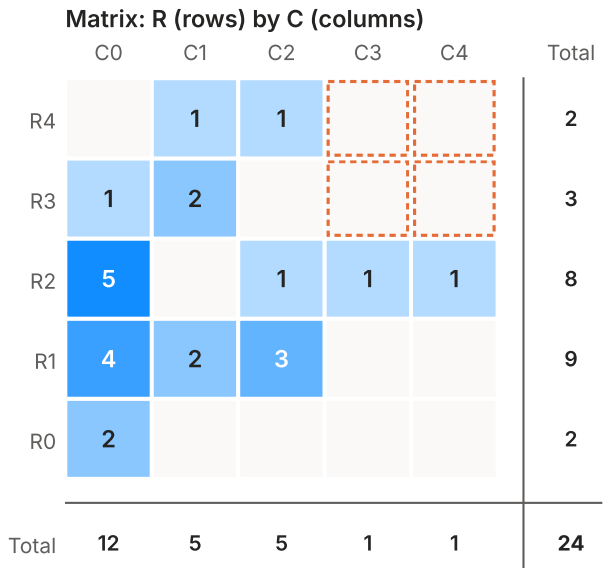


FIG. 2: Matrix of programs by required verification R (rows) and public results C (columns), with row and column totals. Cell shading scales with the number of programs. The dashed cells mark the region $R \geq 3$ and $C \geq 3$, which identifies programs that both measure and publish project-level outcomes; it is empty.

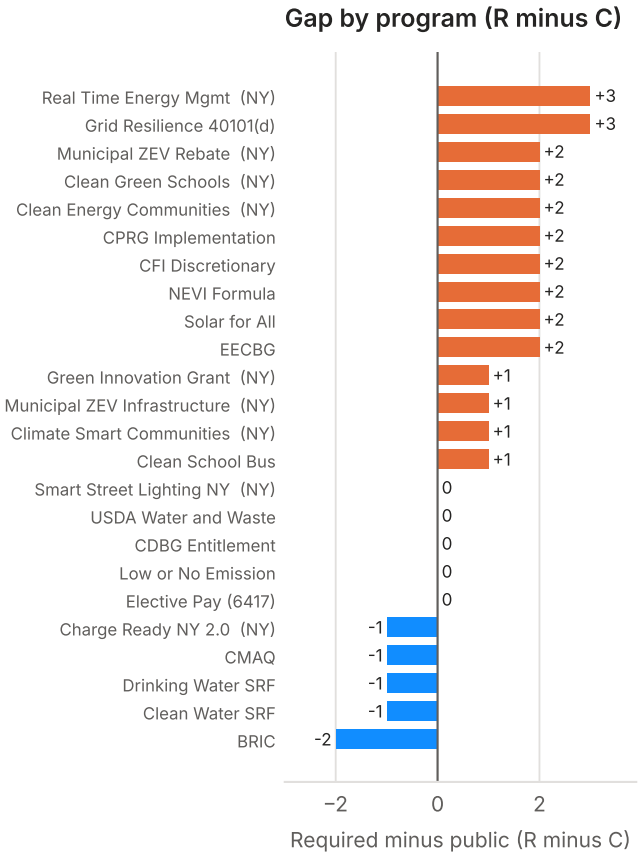


FIG. 3: Required-to-public gap $G_i = R_i - C_i$ for every program, sorted. Positive bars (orange) mark programs whose rules require more verification than they publish; negative bars (blue) mark programs that publish more than their rules require, mostly through agency data portals. NY marks New York State programs.

sured results can diverge by a large margin, and that the divergence is visible only where measurement is required and its results are published.

V. INTERPRETATIONS

A. Sensitivity to low-confidence codes

Because 9 R codes, 5 C codes, and 8 A codes rest on low-confidence evidence, the headline indicators were recomputed under the four treatments of Eqs. (8) and (9). Table VII reports counts, and Fig. 8 shows the same indicators as shares of each sample so that the restricted samples can be compared with the full one.

The central finding is stable. Under every treatment, no program both requires measured or verified outcomes and publishes project-level outcomes. At least 21 of 24 programs in the full-sample treatments (and 9 of 10 in the most restricted sample) remain at or below a modeled baseline for attributability. The share of programs requiring outputs or less is the least stable indicator: across the full-sample treatments it ranges from 6 to 15 of 24, depending on how low-confidence R codes are treated. Of the 9 low-confidence R codes, 6 belong to New York State programs whose reporting rules are set in requests for applications or award contracts not reviewed for version 0.1.0; those programs are the priority for review in version 0.5.

B. Illustrative scenario: disclosure of collected outcome data

To show how the Ledger can be used to evaluate a policy option, consider the scenario of Eq. (10), in which every program that already requires measured or independently verified outcomes publishes those outcomes at project level. The scenario changes no program's rules about what recipients must collect. It affects 5 programs: the Grid Resilience formula grants, NEVI, the Charging and Fueling Infrastructure grants, Clean Energy Communities, and Real Time Energy Management.

Under the scenario (Table VIII, Fig. 9), the number of programs that both measure and publish project-level outcomes rises from 0 to 5, the number with a positive gap falls from 14 to 11, and the mean C code rises from 0.92 to 1.33. Programs with no public results fall from 12 to 11. The attributability distribution is unchanged, because publication does not create a baseline that was not collected. However, two of the five programs, Grid Resilience and Real Time Energy Management, would then publish measured pre and post data that outside

TABLE VII: Headline indicators under alternative treatments of low-confidence codes, Eqs. (8) and (9). Entries are numbers of programs.

Treatment	Programs	$R \leq 1$	$R \geq 3$	$C = 0$	$A \leq 1$	$R \geq 3$ and $C \geq 3$	$G_i > 0$
Base (v0.1.0 codes)	24	11	5	12	22	0	14
(a) Exclude low-confidence R	15	6	5	6	13	0	8
(b) Exclude any low-confidence code	10	5	2	4	9	0	4
(c) Low-confidence codes $+1$	24	6	9	10	21	0	15
(d) Low-confidence codes -1	24	15	5	15	22	0	12

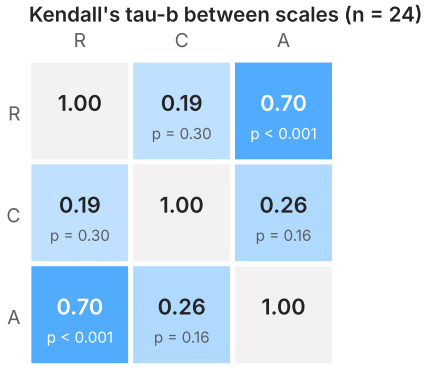


FIG. 4: Kendall's τ_b matrix for the three scales. Shading scales with the coefficient; each off-diagonal cell gives its two-sided p-value.

TABLE VIII: Indicators under current codes and under the disclosure scenario of Eq. (10).

Indicator	Current	Scenario
$R \geq 3$ and $C \geq 3$	0	5
Positive gap $G_i > 0$	14	11
Zero gap	5	8
Negative gap	5	5
$C = 0$	12	11
Mean C	0.92	1.33
$A \leq 1$	22	22
Mean outcome gap $\bar{O}$	2.00	2.00

analysts could pair with comparison sites, which is the step from $A = 2$ to $A = 3$ that no program in the sample has taken.

The scenario also exposes a property of the outcome gap derived in Sec. II: because $O_i = 4 - \max(R_i, C_i)$, it does not change when a program that already measures begins to publish, and the mean stays at 2.00. The outcome gap therefore measures distance from the strongest evidence level on either scale, not alignment between them, and should be read together with the required-to-public gap. Version 1.0 will consider a joint indicator that rewards measured outcomes only when they are also published.

This scenario is illustrative. It is not a recommendation that any agency adopt a specific disclosure rule, and it does not account for confidentiality constraints, such as the anonymization clause in 23 CFR 680.112 or taxpayer confidentiality for elective pay, that may limit

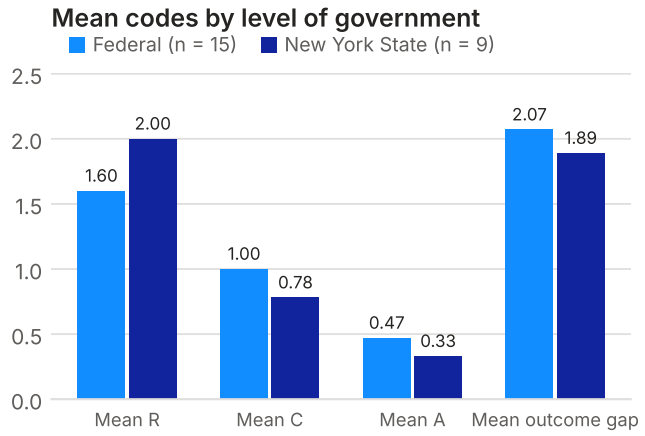


FIG. 5: Mean R, C, and A codes and mean outcome gap $\bar{O}$ for federal and New York State programs.

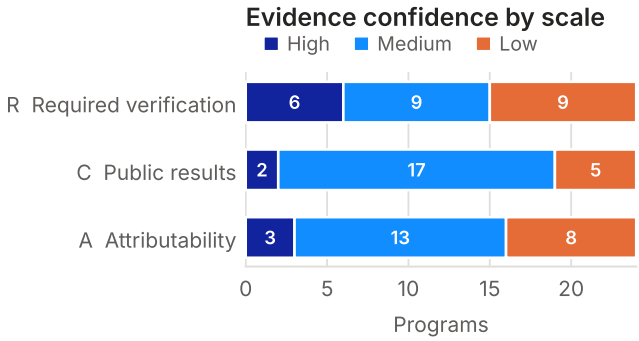


FIG. 6: Confidence of the evidence behind each program's code, by scale.

project-level release.

C. Policy implications

Collection and disclosure are separate decisions. The strength of a program's verification requirements says little about what it discloses. Programs that require metered or independently verified outcomes keep those data internal or release them in aggregate, while the programs with the most open project-level data systems publish modeled estimates. For municipal officials, a program's published figures cannot be read as evidence of realized performance without checking how they were produced. For oversight bodies, a finding of weak disclosure does not

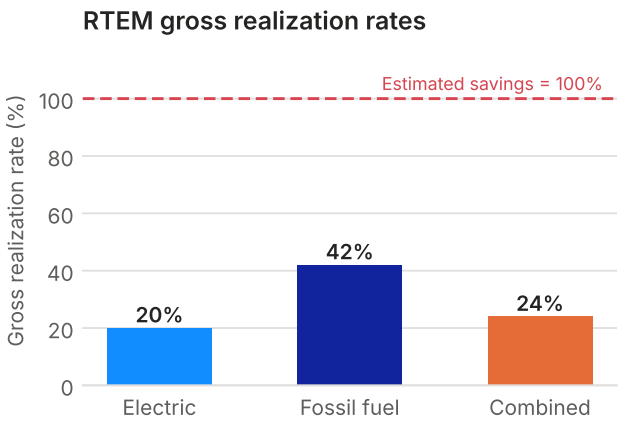


FIG. 7: Gross realization rates ρ reported by the independent evaluation of NYSERDA's Real Time Energy Management program. The dashed line marks full realization of estimated savings ($\rho = 100$ percent).

imply weak collection, and the reverse.

The federal floor permits output-only reporting. In the sample, 11 programs require no more than output reporting, consistent with a Uniform Guidance that recommends rather than requires baselines and outcome indicators. Agencies that wish to support later evaluation can specify baseline and outcome data in award terms, as DOE did in the Grid Resilience metrics guidance [23] and FHWA did in 23 CFR 680.112 [28]. These cases show that measurement requirements can be written into program rules for municipal recipients without an evaluation contract.

Attribution requires design choices made before funding. No program is coded $A = 3$, and only 2 support measured pre and post comparison. Attribution cannot be added after the fact if baselines were not collected. Where programs already require metered operating data, publishing those data at project level with appropriate anonymization would allow independent analysts to construct comparison groups. Where programs fund equipment with predictable energy profiles, such as street lighting conversions or fleet replacement, a simple pre and post consumption requirement would move a program from $A = 0$ to $A = 2$.

Municipal decision support. Municipalities choosing among funding sources typically compare eligibility, match requirements, and award size. The Ledger adds a further dimension: the evidence a program will generate. A municipality that wants to demonstrate realized savings to its governing body, bond investors, or residents may prefer programs that require measured outcomes, or may collect measured data voluntarily where rules do not require it. Because the Ledger is open and versioned, its codes can be incorporated into municipal planning and decision-support tools.

Evaluation and oversight. The Ledger identifies where independent evidence exists and where it does not. Programs coded $R = 2$ with $C \geq 3$, which publish project-level estimates, are candidates for retrospective validation against utility or operating data. Programs coded $R \geq 3$ with $C \leq 1$ are candidates for requests to release existing measured data. Programs coded $R \leq 1$ cannot be validated from program records at all.

VI. LIMITATIONS

The findings should be read with the following limitations.

1. *Publication, not collection.* C measures what is public. Agencies may hold outcome data they do not release, and $C = 0$ means no public results were found through official channels, not that none exist.
2. *Purposive sample.* The 24 programs were chosen to span agencies, funding vehicles, and asset classes. They are not a random or complete sample, and the distributions describe these programs only.
3. *Single coder.* Every code was assigned by one coder and confirmed by the maintainer against its cited source. Inter-coder agreement will first be reported in version 0.5.
4. *Secondary official material.* Many program pages defer reporting rules to award terms or application packages that were not reviewed. These codes carry low confidence, and Sec. A shows which findings depend on them.
5. *Changing requirements.* Several programs were suspended, rescinded, terminated, or litigated in 2025 and 2026. Codes reflect sources accessed in October 2026.
6. *Ordinal scales.* The derived measures of Eqs. (2) and (3) subtract ordinal codes for convenience, and means of codes are summaries only.
7. *Design, not performance.* The Ledger rates what programs require and publish. It does not judge whether any project saved energy or reduced emissions.
8. *Tax credits at the mechanism level.* Elective pay is coded as a single mechanism, although production-based credits use metered output and investment-based credits are cost based.
9. *Manual retrieval.* Sources were read through a web reader and logged by hand with URL and access date, not retrieved by scripted download with checksums, and some may move or change after the access date.
10. *Documentation counts.* The version 0.1.0 codebook lists 79 evidence rows and 4 reference sources, while the released tables contain 80 evidence rows and 3 reference sources. This paper uses the released tables; the codebook will be corrected in the next release.

VII. CONCLUDING REMARKS

Public programs funding municipal energy and infrastructure differ widely in the evidence they generate, and those differences have not previously been recorded in

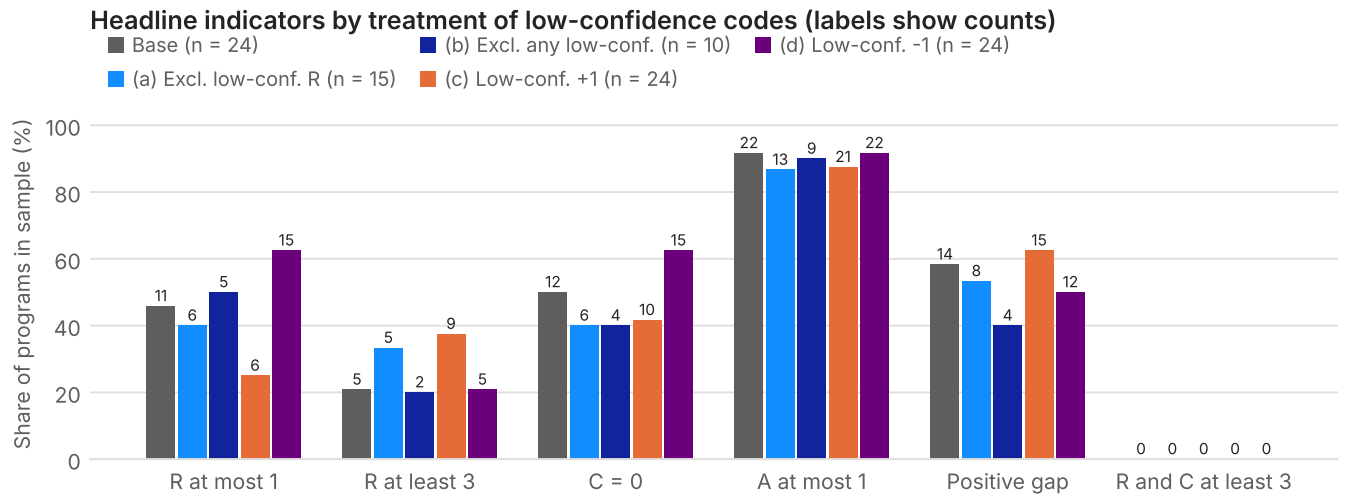


FIG. 8: Headline indicators under the base codes and the four treatments of low-confidence codes, as shares of each sample; labels give counts. The joint indicator $R \geq 3$ and $C \geq 3$ is zero under every treatment.

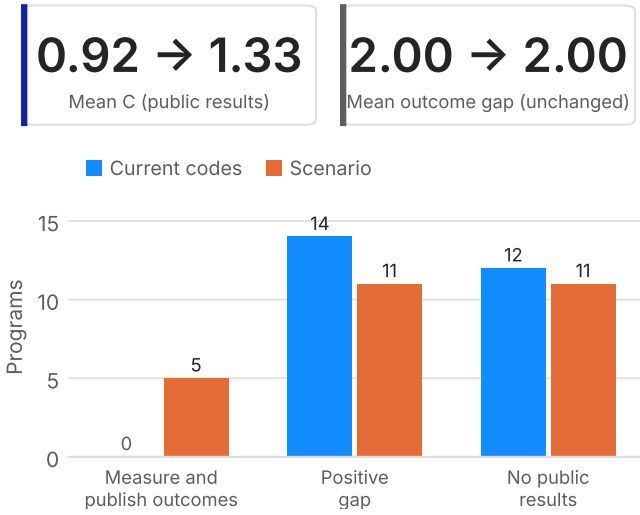


FIG. 9: Disclosure scenario of Eq. (10). Columns compare three indicators under current codes and under the scenario; cards show the change in mean C and the unchanged mean outcome gap.

one public, source-cited register. In this work we introduced the Verification Gap Ledger, which separates three properties that are often conflated: what verification a program requires, what results it publishes, and whether its data could support attribution of outcomes to the funding.

In the first sample of 24 federal and New York State programs, 46 percent require no more than output reporting, 50 percent publish no results beyond award announcements, and 92 percent are coded at or below a modeled baseline for attributability. No program both requires measured or independently verified outcomes and publishes them at project level, and this finding holds under every sensitivity treatment tested. The programs that publish project-level outcomes publish modeled estimates, and the one program with a published billing-analysis evaluation measured electric savings at 20 percent of estimates. Required verification and disclosure are only weakly associated ($\tau_b = 0.19$). Together these results indicate that, for most programs in the sample, the realized effects of public spending on municipal energy and infrastructure cannot be confirmed from public records.

The Ledger is designed to grow and to be tested. Version 0.5 (February 2027) will review award terms for low-confidence codes and introduce an independent second coder on a random 20 percent sample, reporting percent agreement and weighted kappa. Version 1.0 (May 2027) will expand program coverage and refine the derived indicators, including a joint measured-and-published indicator motivated by Sec. B. Version 2.0 (December 2027) will extend coverage further and maintain the register as a searchable public resource. Every code change is recorded with its reason, and identifiers remain permanent. Researchers, program staff, and municipal officials who identify a program that should be added or a code that should be reconsidered can submit the source through the public repository.

DATA AND CODE AVAILABILITY

All data, documentation, and code are openly available. Version 0.1.0 of the Verification Gap Ledger is archived on Zenodo at doi:10.5281/zenodo.23302898; the concept DOI 10.5281/zenodo.23302897 resolves to the latest version. The source repository is https://github.com/Olumide-Akindipe/verification-gap-ledger, and the Ledger can be browsed and searched at https://olumide-akindipe.github.io/verification-gap-ledger.

Every number, table, and figure in this paper is produced from vgl_programs.csv and vgl_evidence.csv by the repository scripts 01_build.py, 05_preprint_stats.py, and 06_preprint_assets.py, and recorded in stats.json and stats_preprint.json. Data and documents are licensed CC BY 4.0; code is licensed MIT.

DECLARATIONS

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APPENDIX A: CODES FOR ALL PROGRAMS

TABLE A1: Verification Gap Ledger v0.1.0 codes for all 24 programs. G and O are the gap and outcome gap of Eqs. (2) and (3). Confidence is listed in $R/C/A$ order (H high, M medium, L low). Ev. is the number of evidence items. Program-level sources for every code are listed with URL, locator, and access date in vgl_evidence.csv.

Ledger ID	Program	R	C	A	G	O	Conf.	Ev.
VGL-F-001	Energy Efficiency and Conservation Block Grant Program (IIJA)	2	0	0	+2	2	H/M/M	4
VGL-F-002	Grid Resilience State and Tribal Formula Grants (IIJA 40101(d))	3	0	2	+3	1	M/L/M	3
VGL-F-003	Clean Water State Revolving Fund	1	2	0	−1	2	M/M/M	3
VGL-F-004	Drinking Water State Revolving Fund	1	2	0	−1	2	L/M/L	3
VGL-F-005	Solar for All (Greenhouse Gas Reduction Fund)	2	0	0	+2	2	L/M/L	5
VGL-F-006	Congestion Mitigation and Air Quality Improvement Program	2	3	1	−1	1	M/M/M	3
VGL-F-007	National Electric Vehicle Infrastructure Formula Program	3	1	1	+2	1	H/L/M	4
VGL-F-008	Charging and Fueling Infrastructure Discretionary Grant Program	3	1	1	+2	1	M/L/M	3
VGL-F-009	Elective Pay for clean energy tax credits (IRA sec. 6417)	0	0	0	0	4	M/H/H	4
VGL-F-010	Low or No Emission Vehicle Program	1	1	0	0	3	M/L/M	3
VGL-F-011	Clean School Bus Program	1	0	0	+1	3	H/M/M	3
VGL-F-012	Building Resilient Infrastructure and Communities	2	4	1	−2	0	M/H/M	4
VGL-F-013	Community Development Block Grant, Entitlement Communities	1	1	0	0	3	M/M/M	3
VGL-F-014	Water and Waste Disposal Loan and Grant Program	0	0	0	0	4	M/M/M	3
VGL-F-015	Climate Pollution Reduction Grants, Implementation	2	0	1	+2	2	L/L/L	4
VGL-NY-001	Clean Energy Communities	4	2	1	+2	0	H/M/H	3
VGL-NY-002	Real Time Energy Management	4	1	2	+3	0	H/M/H	3
VGL-NY-003	Clean Green Schools Initiative	2	0	0	+2	2	L/M/L	3
VGL-NY-004	Charge Ready NY 2.0	1	2	0	−1	2	L/M/M	3
VGL-NY-005	Climate Smart Communities Grant Program	1	0	0	+1	3	L/M/L	3
VGL-NY-006	Municipal Zero-Emission Vehicle Rebate Program	2	0	0	+2	2	H/M/L	4
VGL-NY-007	Municipal Zero-Emission Vehicle Infrastructure Grant Program	1	0	0	+1	3	L/M/L	3
VGL-NY-008	Smart Street Lighting NY	2	2	0	0	2	L/M/M	3
VGL-NY-009	Green Innovation Grant Program	1	0	0	+1	3	L/M/L	3